

CUSTOMER CHURN PREDICTION USING ARTIFICIAL INTELLIGENCE

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Project Title:

Customer Churn Prediction using Artificial Intelligence

Objective:

The objective of this project is to predict whether a customer will leave a telecom company using Artificial Intelligence and Machine Learning techniques based on customer demographic and service usage data.

Tools and Technologies:

- Python
- Google Colab
- Scikit-learn
- Pandas
- NumPy

Dataset:

The Telco Customer Churn dataset was used for this project. It contains customer details such as contract type, tenure, monthly charges, internet service, and churn status.

Methodology:

Loaded the dataset into Python environment.

Performed data preprocessing and cleaning.

Removed irrelevant columns (Customer ID).

Converted categorical variables into numerical values using Label Encoding.

Handled missing values in TotalCharges column.

Split dataset into training and testing sets.

Trained a Random Forest Classifier model.

Predicted customer churn.

Evaluated model performance using accuracy and classification report.

Output:

- Accuracy: Approximately 80%
- Precision, Recall, F1-score: Evaluated for both classes
- Predicted Result: Customer Will Churn

Result:

The model successfully predicted customer churn with an accuracy of approximately 80%. The model performed better in identifying customers who stay compared to customers who churn.

Conclusion:

This project demonstrates how Artificial Intelligence can be used in the telecom industry to predict customer churn. The Random Forest Classifier provided good accuracy and useful insights for customer retention strategies.